DeCad

Verified engineering report

23 October 2026 | Application b0d6c9b | Evidence cff361e | Recovery 871ee61

The review-assisted application is verified. Automatic schematic recovery remains incomplete. This report separates accepted graph improvements, rejected experiments, serialization fidelity and manual corrections.

Measurement	Before	Accepted
Clear exact assignments	227/245	227/245 (92.65%)
Native valid assigned pins	103	103
Native correct connected pairs	9	9
Native incorrect connected pairs	14	13
Native pair precision	39.13%	40.91% (+1.78 points)
Native CAD readback	18/18 each	18/18 each; 2,206 pins each

Accepted improvement: Wio R1.1--R1.4 is no longer connected through an incompatible OCR reading. Invalid observations R1.4 and R1.55 remain available as review evidence. No replacement number or signal label is inferred.

Final verification: 384 backend tests passed, 6 environment-dependent skipped; 65 focused tests and 18 frontend tests passed. All previously correct native pins, source labels and connected pairs were retained.

Remaining automatic errors: 13 native false pairs and all 18 clear errors. Native stress-corpus pair recall remains 9/21,025 (0.0428%). Dense native targets can include hidden or unprinted information; the result does not certify whole-board automatic recovery.

Changes, preservation and evidence boundaries

Only the passive-pin guard was accepted. The explicit two-terminal R/C/L/D library type must be corroborated by observed terminals 1 and 2 with collinear, opposed outward anchors. Extra IDs are quarantined outside the electrical graph; original nodes, previous nets, anchors, warnings and review regions survive canonical review serialization.

Changed application module: backend/app/local_recovery.py. Tests include malformed-number and uncertain-identity cases, three opposing-anchor rejection cases, and an audited real Wio excerpt checked through canonical pin inventory and an independent XML reader.

Rejected parser experiment: a stricter connector boundary removed a phantom pair but created nonexistent P3.2 on another design through regex reference backtracking. Execution stopped. digital_vision.py was restored byte-for-byte to the accepted baseline. That apparent improvement is not shipped.

Accepted evidence mode: exact guard replay over all 18 baseline graphs, with no graph changes on the other 17, plus fresh current-code Wio and ESP32 rollback workers. Sixteen final cohort outputs use unchanged accepted recognition and two use current fresh targets. Rejected candidate fresh workers are archived separately and never relabeled as current acceptance.

Preserved achievements: TD9-style XML codec and inventory; persistent canonical review drafts; revision checks and saved undo; manual component and net correction; split/merge; confidence handling; stale-source invalidation; exact observed-pin native exports; frontend; and launcher readiness checks.

Independent faculty XML checks retain all 129 nodes, 41 nets and meaningful attributes: 114 used / 15 unused pins, four QD0 members, and used-but-unassigned J2.15. Actual vendor TD9 import remains NOT VERIFIED.

No training, large model download, new dependency or application UI redesign. The protected OCR model is unchanged:

c5ad0e9ebe43373f1986ed5d6c39be0b73a29cb4dcb257e45ca7f91f41ce07cf

Visual UI interaction was disabled by Work settings and was not repeated. Live HTTP upload, recognition, saved-review reopening and downloads passed. Existing frontend regressions pass.

All 14 original native false pairs

Case	Pair	Independent reference	Status
Grove U2	R34.1 -- U2.3	GND / VBUS	OPEN
Grove U2	R34.1 -- U2.6	GND / V33IO	OPEN
Grove U2	R34.1 -- U2.9	GND / VBUS	OPEN
Grove U2	U2.17 -- U2.3	GND / VBUS	OPEN
Grove U2	U2.17 -- U2.6	GND / V33IO	OPEN
Grove U2	U2.17 -- U2.9	GND / VBUS	OPEN
Grove U2	U2.2 -- U2.3	GND / VBUS	OPEN
Grove U2	U2.2 -- U2.6	GND / V33IO	OPEN
Grove U2	U2.2 -- U2.9	GND / VBUS	OPEN
Grove U2	U2.3 -- U2.6	VBUS / V33IO	OPEN
Grove U2	U2.6 -- U2.9	V33IO / VBUS	OPEN
Wio R1	R1.1 -- R1.4	Pin 4 does not belong to this two-terminal resistor	REMOVED
nRF GPIO	P0.00 -- P0.31	P0 is not a physical component	OPEN
CM4 U6	U504.4 -- U504.9	Source owner is U6, not U504	OPEN

Grove pins 3/9 are VBUS and pin 6 is V33IO, while pins 2/17 and R34.1 are genuine ground. Capacitors and unresolved anchors were mistaken for shared ground paths. U2.3--U2.9 is a previously correct pair even though its detected net name is wrong; a guessed split must not destroy it.

CM4 U6.4--U6.9 is a real junction. The error is the fused reference U504, not a false physical dot. Source images follow on later pages.

All 18 clear-sample errors

These are 11 missing physical-anchor observations and 7 incorrect assignments. Printed source labels can be read manually; no sample-specific substitutions or unverified recovery connections were inserted.

Sample	Pin	Observed net	Expected net	Remaining cause
S1	U64.4	WZPLJ6	DD6	Wrong label association or unresolved local trace
S1	U64.13	missing	CC7	Missing physical anchor; fused or crowded number/label strip
S3	U6.14	KAAI	AAA1	Wrong label association or unresolved local trace
S3	U6.11	missing	AAA4	Missing physical anchor; fused or crowded number/label strip
S3	U8.14	AAA6	AAA9	Wrong label association or unresolved local trace
S3	U8.11	missing	AAA12	Missing physical anchor; fused or crowded number/label strip
S3	U27.17	missing	AR12	Missing physical anchor; fused or crowded number/label strip
S3	U27.7	RG9	AR3	Wrong label association or unresolved local trace
S3	U28.19	missing	AR10	Missing physical anchor; fused or crowded number/label strip
S4	U31.27	missing	ADDA11	Missing physical anchor; fused or crowded number/label strip
S4	U33.14	missing	ADDA9	Missing physical anchor; fused or crowded number/label strip
S4	U34.9	missing	ADDA7	Missing physical anchor; fused or crowded number/label strip
S5	U36.21	missing	INDA14	Missing physical anchor; fused or crowded number/label strip
S5	U36.2	N00549	INDA12	Wrong label association or unresolved local trace
S5	U49.2	~TT2	INDA8	Wrong label association or unresolved local trace
S5	U49.9	N00619	INBU8	Wrong label association or unresolved local trace
S5	U37.9	missing	INBU4	Missing physical anchor; fused or crowded number/label strip
S5	U50.9	missing	INBU0	Missing physical anchor; fused or crowded number/label strip

Low-resolution limits remain: the 200-dpi scan and blurred cases have 0/72 exact assignments. The 300-dpi scan retains 44/72. Clear results remain S1 54/56, S3 62/69, S4 69/72, S5 42/48.

Why 11 faculty exports still need review

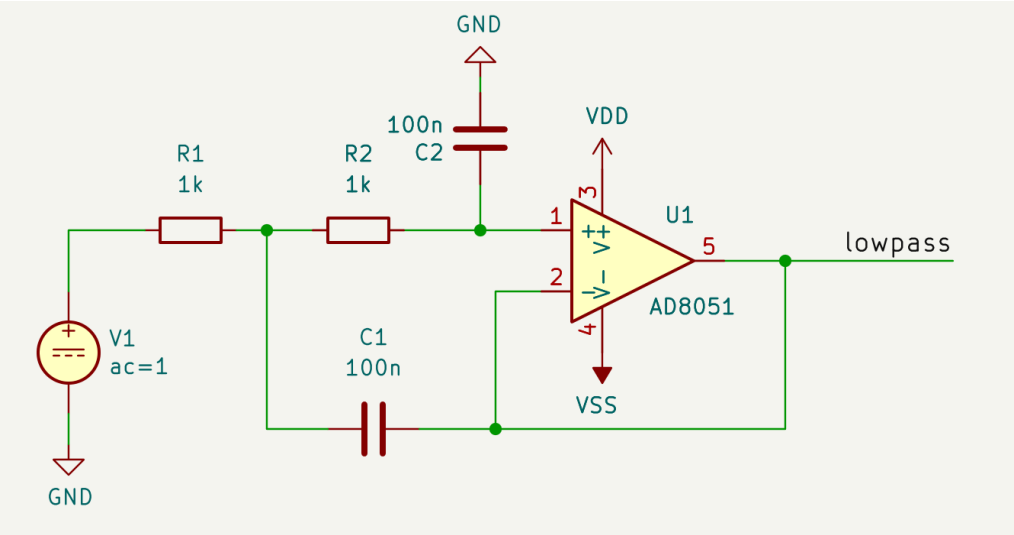
All observed pins and assignments round-trip. The 11 blocked graphs contain 37 component records with no observed physical IDs. A component-only faculty node would require an invented pin. No codec defect was found.

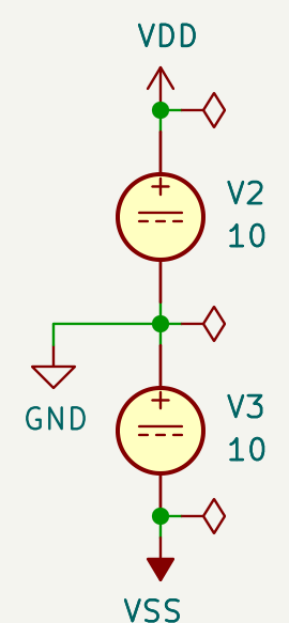
Case	Zero-pin component records	Source finding
S3	R1, R6	R1 aliases printed R11; R6 is visible, numbering unprinted.
S4	R11, R13, R20	Visible resistor bodies; physical numbers unprinted.
S5	R12, R19	Visible resistor bodies; physical numbers unprinted.
S4-scan300	R1, R11, R14, R15	R1 aliases R13; other resistors visible with degraded ink.
S4-scan200	R11	Blurred but visible R11; numbers unprinted.
Grove Vision	C36, R24, R32, R40	Visible bodies; R32 prints 1/2. C36 value also misassociated.
XIAO ESP32	R11, R15, R2, U4, U58	Visible resistors; U4 from prose and U58 from USB text.
AU7 native	R1, R2, R8, R9	Visible resistors with unprinted numbers.
Raccoon native	R3	Visible resistor; numbers unprinted.
PL test native	R1, R2, R4, R5, R7, R8	Visible resistors; duplicate references also occur in a component list.
CM4 native	R5, R6, R8, R9, U4	Resistors omit numbers; U4 visibly prints 1..5 but has no extracted anchors.

Unprinted physical numbering is a genuine source-information limit. A visible symbol with missed anchors is a recognition defect. Prose and fused-reference false components need component review. These must not be conflated with exporter defects.

Unseen Sallen-Key: raw versus reviewed

Unchanged installed KiCad demonstration; raster only supplied to recognition. Raw output was saved before the native reference was read. The source is shown in two crops from the same image: signal circuit and remote supply blocks.



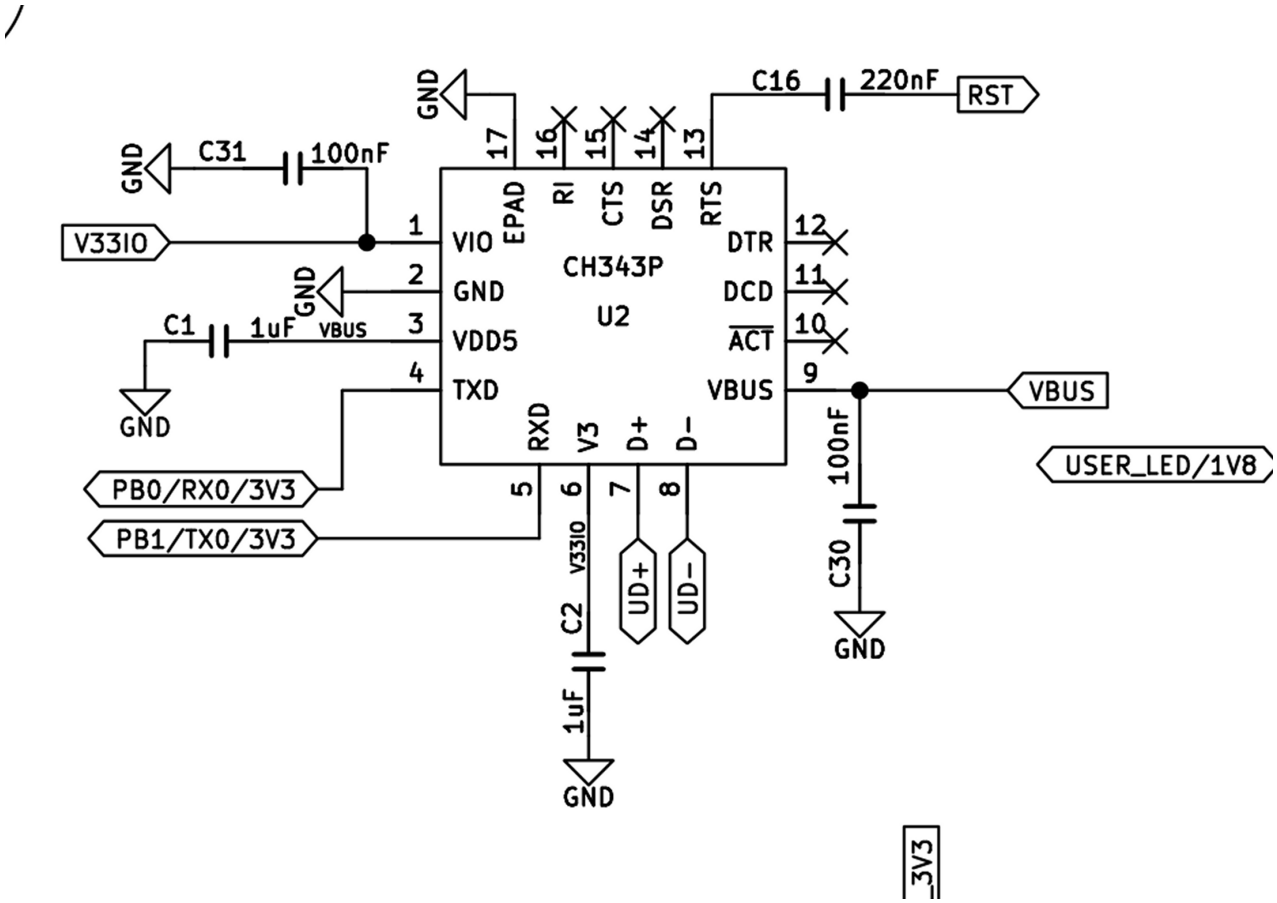


Measure	Raw recognition	Manual review + native pin lookup
Components	2/8	8/8, values exact
Assigned physical pins	0/19	19/19
Positive pairs	0/18	18/18
False pairs	0	0
Negative pairs preserved	Not a complete graph	153/153

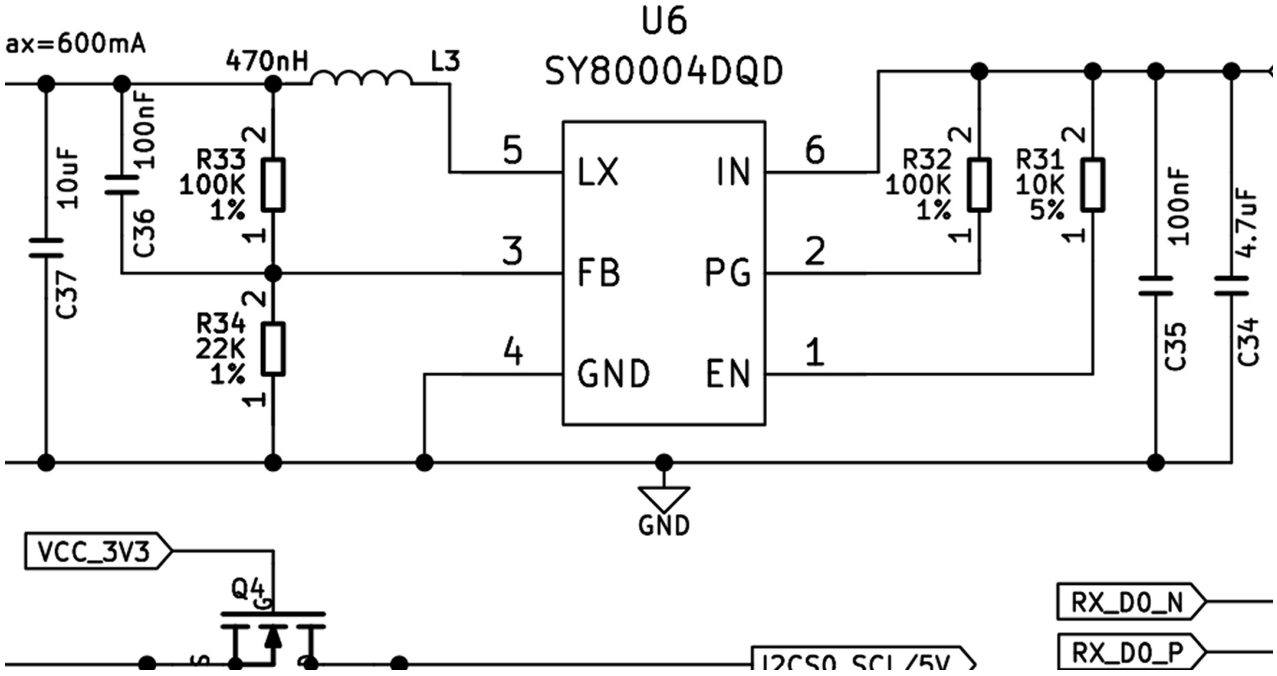
The first manual convention produced 6 correct and 12 wrong pairs. The raster omits R/C terminal IDs; native embedded symbol pins and rotations resolve them. R1/R2/C1 pin 1 is at the right; C2 pin 1 is below. The initial transcription and failed comparison remain in the evidence.

This is not an automatic accuracy gain. Reviewed XML and both native readbacks preserve all 19 assignments with zero extras. All 11 named-label assignments match after root-sheet slash normalization; literal full names match 8/19 because unprinted intermediate nets use different generated names.

Source evidence: Grove ground propagation

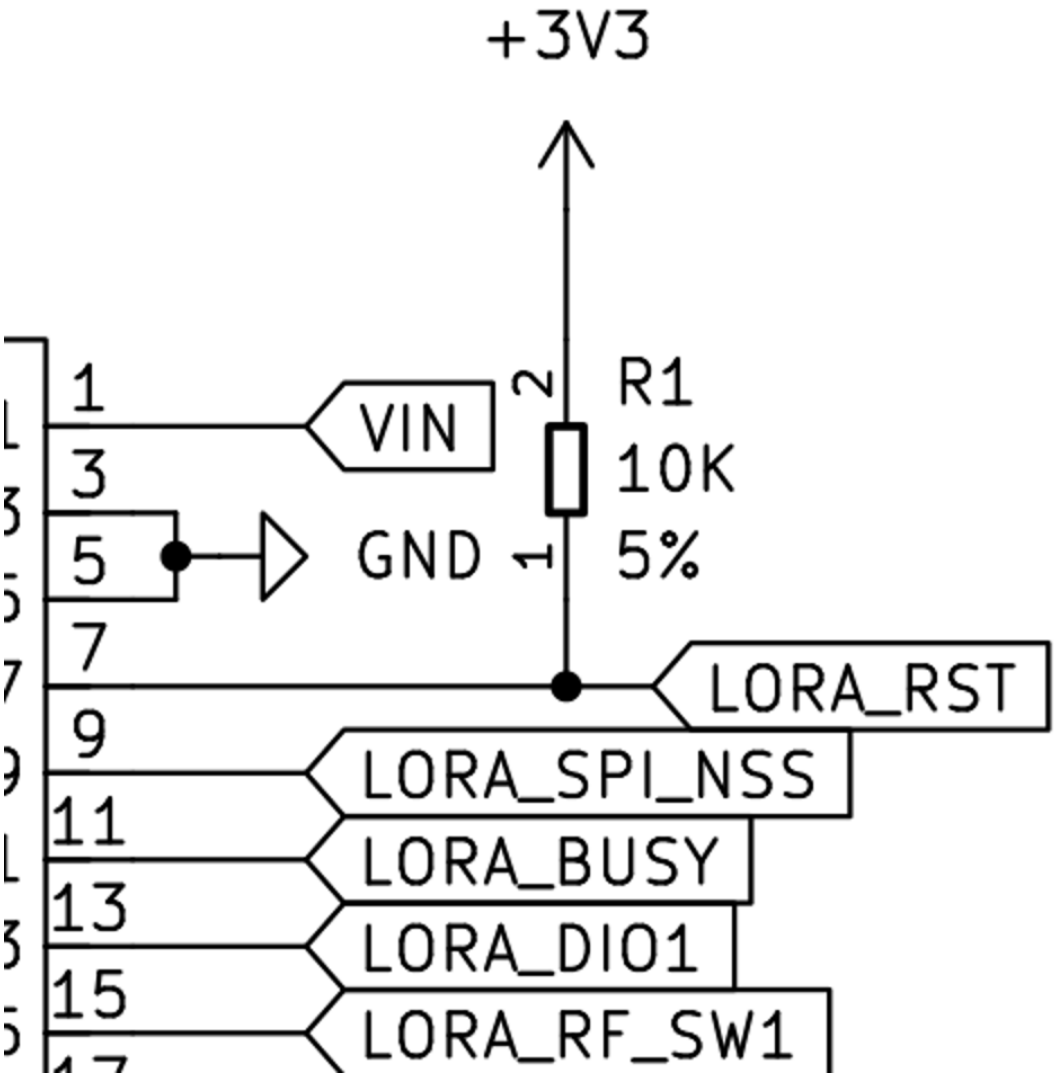


U2 pins 3/9 are VBUS and pin 6 is V33IO. C1/C2/C30 separate these nets from ground. The accepted pass does not guess a global rail split.



Separate source region: R34 pin 1 visibly reaches the ground rail. R34.1 with U2.2/U2.17 is a valid ground relationship to preserve.

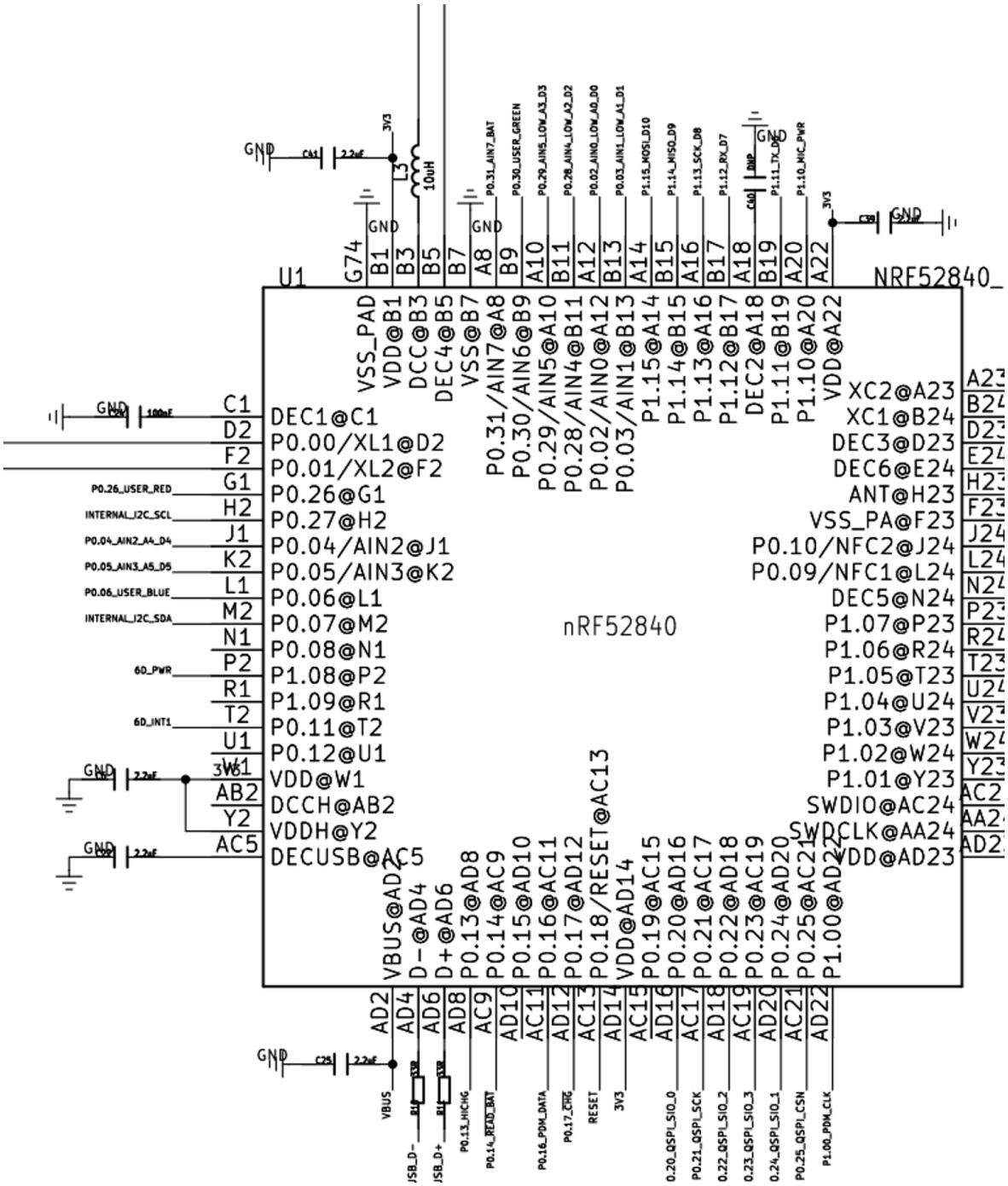
Source evidence: accepted Wio correction



R1 is a 10K resistor with printed terminals 1 and 2. The raw R1.4 and R1.55 observations cannot be terminals of this corroborated two-terminal symbol. The guard quarantines them without assigning either to another component or inventing the missing LORA_RST label.

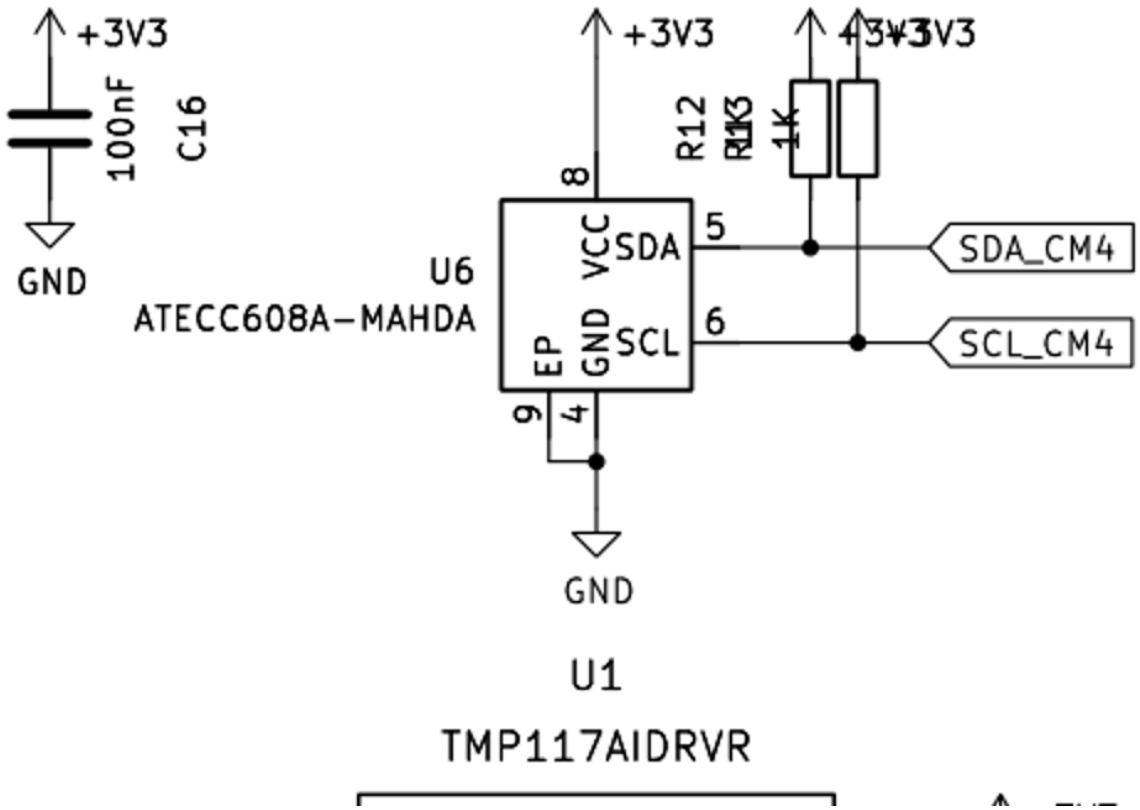
Fresh current-code result: 17 previously valid pins and the correct pair retained; R1.1--R1.4 removed. Every rejected reading remains in source-review metadata. Canonical inventory and independent XML tests prove these readings do not reappear as exported physical nodes.

Source evidence: unresolved nRF phantom port



P0.00/XL1 and P0.31/AIN7 are U1 function names. They are not a separate physical P0 connector. The parser experiment that removed this pair regressed elsewhere, so it was rejected. This false pair remains open in the accepted application.

Source evidence: CM4 owner versus junction



The source reference is U6 (ATECC608A-MAHDA). Pins 4 and 9 visibly join with a dot and reach ground. The detected U504 is fused internal text. A junction-cut repair would damage real connectivity; component identity needs correction first.

No speculative body rewrite or source-specific reference replacement was merged. This error stays explicit in the remaining-pair inventory.

Launch, reproduce and independently review

Repository: D:\DeCad-main\Jrk_DeCad\Schematic2SDN

Run: & .\start-native.ps1

Workbench: http://localhost:3100; API: http://localhost:8100

Choose PrecisionUnseenFilter, then Saved review. Its source raster is attached.

Recheck existing review: backend/.venv/Scripts/python.exe tools/eval/precision_demo.py --verify

Recovery branch: recovery/precision-871ee61; accepted application commit: b0d6c9b.

Reproduction scripts: precision_acceptance.py checks the exact 18-graph guard delta with two current fresh target workers; precision_audit.py recreates source crops and independent inventories. Full commands and evidence paths are in training/TONIGHT_FINAL_ACCEPTANCE.md.

Verified implementation history

Commit	Verified change
cff361e	Record source audits, rejected parser regression and verified precision acceptance
b0d6c9b	Quarantine corroborated passive pin conflicts without inferring connections
871ee61	Document verified tonight review workflow and final acceptance evidence
c6763c1	Flag observed pin conflicts against explicit passive symbols for review
1cc91b8	Invalidate source traces on component edits and manual net splits
eb4f5f3	Add alias scope and stale overlay acceptance checks
0a078d9	Measure final gate resources with Windows APIs and no added dependency
575540a	Invalidate stale trace overlays after manual graph edits
c6f224c	Wait for API readiness before opening the native workbench
40ddbf7	Keep active recognition identical with or without Capture installed
25d6af6	Complete editable review XML workflow and flag shorts without guessed edge cuts
17a921e	Add faculty XML graph interchange and persistent reviewed drafts

Final limits: 13 native false pairs, 18 clear errors, incomplete automatic unseen recovery, unprinted physical-ID ambiguity, no repeated visual UI run, and TD9 vendor import not verified. Serialization fidelity is not evidence of image-recognition correctness.